

A Supplementary Materials

This document contains supplementary materials for our submission “ERASE – A Real-World Aligned Benchmark for Unlearning in Recommender Systems”. *Note that at the time of submission, some SGL model checkpoint files were lost. The corresponding experimental logs are included. We are currently recomputing the lost model files, and will upload them as soon as the re-runs finish.* We report the performance of popularity-based, random, and neighborhood-based baseline methods in Table 15, Table 16, and Table 17. This comparison demonstrates that the additional computational cost associated with neural models is justified by their improved utility.

A.1 Necessary Modifications to the Unlearning Algorithms

Several existing unlearning algorithms require minor but important implementation adaptations to ensure numerical stability, computational efficiency, and fair comparison. In our codebase, unlearning is orchestrated in `recbole/quick_start/quick_start.py` (construction of per-step *forget/clean-forget/retain* loaders, exclusion of previously unlearned users, and retain subsampling) and dispatched/implemented in `recbole/trainer/trainer.py` (method-specific update rules). A shared practical consideration across methods is that we do not use the full retain set per unlearning step. Instead, we construct a per-step retain loader by sampling complete user sessions from the remaining training data while excluding users already unlearned in earlier steps, and we cap the total number of retain interactions per step (additionally bounded by a fixed fraction of the full training interactions). This keeps runtime comparable across methods while maintaining consistent access to retain information.

Stability safeguards are shared across multiple second-order and influence-function-style methods. In particular, the optional hyperparameter `max_norm` acts as a general-purpose safeguard: it can clip the magnitude of the computed parameter update (SCIF, GIF, CEU, IDEA) or clip gradients during repair fine-tuning (Seif). In addition, the implementations include explicit NaN/Inf checks and early stopping of iterative solvers when divergence is detected.

CEU. Our CEU implementation follows the certified unlearning structure and introduces numerical stabilizers directly in the second-order computations. The update uses an L_2 -type regularization term controlled by `ceu_lambda` when forming the relevant Hessian(-vector) quantities, and it also adds a corresponding damping term in the Hessian-vector product computation. We apply these stabilizers during the unlearning update (not as a requirement of the initial training objective), because they substantially improve robustness of the conjugate-gradient solve and reduce divergence. Before adding certification noise with standard deviation `ceu_sigma`, we optionally clip the influence estimate using `max_norm`.

Fanchuan. For Fanchuan, both the contrastive-learning representation and retain usage are critical. We fix the contrastive representation (via `Trainer.get_embedding_for_contrastive_learning`) as follows: user embeddings for CF (when the model exposes `user_embedding`), sequence/session representations for SBR (via the sequential forward pass), and basket-level representations for NBR (via `get_basket_embedding` when available, otherwise a forward-pass fallback). To make the method feasible under sequential unlearning, we avoid full-retain repair rounds: contrastive updates draw retain batches from a shuffled index pool over the per-step retain loader, and the subsequent retain round operates on a bounded number of retain samples per unlearning call (with epochs adjusted if the requested sample budget exceeds the unique samples in the retain loader).

GIF. GIF does not require structural changes, but it is highly sensitive to numerical instability in the Hessian-inverse approximation. Our implementation uses a Neumann-series-style approximation with damping (`gif_damping` and `gif_scale_factor`), includes NaN/Inf checks throughout Hessian-vector product computation and iteration steps, and can optionally clip the final update with `max_norm`. Despite these safeguards, GIF remains among the methods that diverge most frequently in practice.

IDEA. IDEA similarly benefits from stability safeguards. Our implementation solves $(H + \lambda I)x = v$ with damping `idea_damping`, includes NaN/Inf and non-positive-definite checks in the iterative procedure, optionally clips the update (`max_norm`) before computing certification quantities, and applies the certified noise mechanism (with `idea_sigma` increased if needed to satisfy the (ϵ, δ) requirement).

Kookmin. For Kookmin, the central hyperparameter is the per-layer reset ratio, implemented as `kookmin_init_rate`. During each unlearning step, we reinitialize a fraction of parameters *within each tensor* based on the signed-gradient signal (using a per-layer top- k thresholding rule), then perform a short retain round whose effective workload is scaled to the forget set size. Additionally, in `quick_start.py`, we reduce the base learning rate when running Kookmin unlearning to prevent instability when only a subset of parameters is reset. We tune `kookmin_init_rate` in our experiments to avoid both collapse from overly aggressive resets and ineffectual updates from overly small resets.

SCIF. In SCIF, a key adaptation is selecting which parameters are updated depending on the task/model. The code updates user embeddings when explicit `user_embedding` exists (typical CF models). For common sequential recommenders, we update a task-appropriate subset (e.g., item embeddings and final/linear layers depending on the architecture), while for models where no special-case mapping is defined (including NBR models such as Sets2Sets) the implementation defaults to updating all trainable parameters. SCIF uses a conjugate-gradient inverse-HVP solve with damping (`damping`) and includes NaN/Inf checks; when recurrent layers are present, we disable cuDNN for the second-order derivative path to make Hessian-vector product computation feasible. The final update can be magnitude-clipped via `max_norm`. We carried out a hyperparameter test and found `max_norm = 10` to be the best choice.

Seif. Seif follows its erase/repair structure in our implementation (noise injection into embedding/linear-like layers followed by a bounded repair phase on retained data) and does not require additional algorithmic changes beyond these practical adaptations; optional `max_norm` gradient clipping is supported during the repair phase.

A.2 Extended Discussion on Preliminary Benchmark Results

Why can unlearning outperform retraining on utility metrics? In sensitive item unlearning for CF we observe that unlearned models match or even exceed the recommendation utility of models retrained from scratch on the retained data (LightGCN, SGL). A plausible explanation is *underfitting of retraining under a fixed compute budget*. In our experiments, training, retraining, and unlearning were all subject to the same maximum wall-clock time of one day. Retraining starts from random initialization (or a cold start) and may not reach the same optimum within the time limit, especially for large datasets or slower models. In contrast, unlearning methods start from the fully trained original model and apply a targeted “forget” update followed by a “repair” update on retained data. Because the repair phase continues optimization from a strong initialization, unlearned models can keep improving during repair while retrained models may still be in an earlier optimization regime. This effect is most likely when retraining is compute-limited; with sufficiently long retraining, we expect the gap to diminish.

Comparative trends across methods. Across our benchmarks, SCIF is the most consistently strong method, often yielding the best overall utility while maintaining good forgetting effectiveness. GIF behaves similarly when converging but typically incurs a modest utility degradation compared to SCIF; however, GIF tends to achieve lower sensitive-item exposure (i.e., better forgetting) than SCIF. Seif performs poorly in several multi-request regimes; we attribute this primarily to the interaction of small unlearning batch sizes with repeated noise injection, where the cumulative perturbation disrupts the model more than the repair step can compensate. Finally, GIF is up to an order of magnitude slower than SCIF, which is consistent with GIF updating a broader set of parameters (and/or relying on more expensive inverse-Hessian approximations) than SCIF’s more selective update strategy.

Interpreting “good” and “bad” unlearning performance. We evaluate unlearning along three axes: *utility* (recommendation quality on retained data), *effectiveness* (degree of forgetting; e.g., sensitive-item exposure), and *efficiency* (time per unlearning request relative to retraining). Following prior work (e.g., CURE4REC [1] and MUBox [2]), we report results relative to retraining and treat retraining as the reference for exact data removal. The literature does not provide universal thresholds that map a relative difference to “acceptable” or “unacceptable” unlearning across datasets and models. Therefore, we interpret outcomes comparatively: an unlearning method is preferable when it (i) is close to retraining in utility, (ii) achieves equal or better forgetting effectiveness, and (iii) is substantially more efficient than retraining. We argue that an unlearning algorithm providing a speedup of ≥ 3 orders of magnitude over retraining is efficient enough for practical usage. From the interviews, we learned that training time usually does not exceed one day, and the unlearning time for 1 request does not exceed a few minutes when assuming this upper bound.

Why do some methods fail or degrade? Failure modes differ by algorithmic mechanism and by the structure of the deletion request. For GNN-based recommenders, inverse-Hessian-style approximations can be numerically fragile; methods that rely on less stable iterative schemes may diverge when the curvature is ill-conditioned or when updates accumulate over many requests. More broadly, methods that inject noise, for certified guarantees or just for forgetting, can suffer from *cumulative perturbation*: repeated noisy updates over many batches can move parameters into regions that require long repair to recover, eroding both utility and efficiency.

Method-specific observations and hypotheses.

- Fanchuan: For NBR sensitive-item unlearning, we observe a sharp utility drop alongside strong forgetting indicators. This suggests that the uniform pseudo-label phase can substantially distort item representations, and the subsequent repair phase may not fully recover from this catastrophic shift under the fixed time budget.
- Kookmin: Kookmin performs well for sensitive-item unlearning, consistent with the hypothesis that only a small parameter subset (e.g., item embeddings) needs to be reset and then repaired. In spam unlearning, performance degrades as the number of unlearning batches increases (NBR: few batches, acceptable; SBR: many batches, poor). A plausible explanation is that resets inadvertently affect parameters tied to popular items that appear in spam sessions; because these items are widely used, even small representation damage can hurt many users, and the repair phase may be insufficient to fully restore them under the compute limit. Also, the attack interweaves target items with popular items within sessions; consequently, harming steps that affect all items in the session can also damage representations of popular items, reducing utility broadly. A targeted variant that applies the harming stage only to identified target items could mitigate this, but it introduces an adversarial consideration: attackers could attempt to exploit the unlearning mechanism to suppress items by submitting interactions that later get removed.
- Seif: Seif is competitive when the number of unlearning batches is small, but degrades markedly with many sequential requests. This is consistent with fixed-noise injection accumulating over batches; the repair step cannot keep pace unless its budget is increased, at which point the benefit over retraining diminishes.
- SCIF: SCIF yields strong utility and robust forgetting across most settings. Its main limitation is runtime on large models/datasets, where inverse-Hessian-style computations (even approximated) become expensive; on the largest configurations we observe only a small efficiency advantage over retraining. This suggests that further curvature approximations or more aggressive parameter selection may be

needed at scale. In spam unlearning, SCIF’s stability may also benefit from combining forgetting and repair into a single update rather than applying them sequentially.

- GIF: GIF often achieves excellent forgetting effectiveness (low sensitive exposure) but tends to reduce utility more than SCIF and can be slower. One explanation is that GIF’s parameter selection via k -hop neighborhoods may update a broader or noisier parameter set than SCIF’s recommender-tailored selection (e.g., focusing on item embeddings while freezing others). In addition, GIF’s Neumann-series inverse-Hessian approximation is sensitive to convergence conditions (e.g., requiring the spectral radius of $(I - \lambda H)$ to be < 1), which may explain occasional instability or divergence for some seeds.
- IDEA: IDEA can exhibit similar degradation patterns to noise-based methods (e.g., Seif) under many requests. Its norm-dependent noise, while motivated by certified guarantees in simpler settings, can still lead to cumulative perturbation and long recovery times in deep recommenders. We also observed practical issues at scale (e.g., memory usage) that limit applicability for large datasets.
- CEU: CEU combines noise with L_2 -regularization in the Hessian computation to encourage positive definiteness. It succeeds for LightGCN but fails for SGL in our experiments, suggesting that the additional self-supervised components and augmentations in SGL yield a more complex optimization landscape where the approximation assumptions break down.

Hyperparameter choice. We report the hyperparameter choices for all compared methods.

SCIF and Other Influence-Function-Based Methods. For SCIF, we varied the gradient clipping threshold

$$\text{max_norm} \in \{10^{-2}, 10^{-1}, 10^0, 10^1\}.$$

For other methods relying on influence-function-based parameter updates (GIF, CEU, and IDEA), we used the same max_norm value for consistency. Table 12 reports the performance of SCIF under different clipping thresholds. We chose $\text{max_norm} = 10$ for our experiments.

Kookmin. For Kookmin, we tuned the reinitialization rate:

$$\text{kookmin_init_rate} \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}\}.$$

Table 13 summarizes the results. We chose $\text{kookmin_init_rate} = 10^{-4}$ for our experiments.

Fanchuan. The Fanchuan method involves several design choices related to contrastive learning. These choices can be viewed as a hyperparameter search over augmentation strategies and loss configurations. Due to space constraints, we describe the selection process and the final configuration in the main paper.

Seif. For Seif, we follow the original paper and sample the noise added to the model parameters from a multivariate Gaussian distribution,

$$\mathcal{N}(0, 0.6^2 I).$$

We did not further tune this variance.

Unlearning batchsize. Unlearning interactions of $n \gg 1$ users at the same time is not realistic for sensitive item unlearning, but sensible for removing spam. Consequently, when spam interactions are identified, they are likely to originate from multiple users rather than a single account and can be merged to one unlearning request as opposed to sequential requests for single users. We test $\text{unlearning_batchsize} \in \{2^i | 0 \leq i < 10\}$ and find 2^8 to yield the best results.

Impact of model architectures. Unlearning *attention-based models* with general methods like Kookmin and Seif work well for a small number of unlearning requests, but the performance of the unlearned model drops the more unlearning requests are served. Fanchuan and SCIF generally maintain effectiveness and utility while being multiple orders of magnitude faster than training, but are sometimes too slow (only 1 order of magnitude faster than training for NowP spam). For the *recurrent model* GRU4Rec, only SCIF maintains most effectiveness and efficiency while being similarly fast as other unlearning algorithms. The general unlearning algorithms Kookmin and Seif degrade performance more than 10% in the spam unlearning scenario while Fanchuan works for spam unlearning within the 5% error margin, but degrades performance to $< 25\%$ of the retrained model in sensitive item unlearning while predicting more sensitive items than the retrained model. In NBR for Sets2Sets, Kookmin performs best, followed by SCIF. Kookmin maintains utility and predicts a similar number of sensitive items as the retrained model, likely due to the small number of unlearning requests compared to SBR or CF. For *graph-based models*, we discuss the three graph-specific unlearning algorithms. These algorithms are efficient across scenarios and tasks and have among the best effectiveness and utility in some cases (CEU NowP SRGNN; GIF, CEU Food LightGCN; GIF, IDEA DNNTSP). However, they are also unstable, as seen in Figure 2 in the main paper, especially GIF and IDEA diverged occasionally during the sequential execution of unlearning requests. Seif degraded performance too much due to noise being too big of a factor, as in every setting with multiple sequential unlearning requests. The recommender system-specific method SCIF maintained the highest utility and effectiveness in most experiments, but was sometimes outperformed by graph-specific algorithms. The general methods Fanchuan and Kookmin are efficient, effective, and maintain utility for CF sensitive item unlearning, but are less effective than SCIF in spam unlearning.

A.3 Additional Details for “A Snapshot of Current Progress in Machine Unlearning for Recommender Systems”

Unlearning effectiveness. For unlearning spam, we observe that Fanchuan is volatile with results having effectiveness close to the retrained model but also results with performance up to 22.8% worse than the retrained model. Seif diverges likely due to noise being

too oppressive. Kookmin is often close to the retrained model or more effective for sensitive item unlearning, but in SBR spam unlearning, degrades effectiveness by 15 – 20%. The graph-specific unlearning algorithms diverge often after many unlearning requests, but are close to the effectiveness of the retrained model in NBR spam unlearning and CF sensitive item unlearning. The recommender system specific method SCIF has an effectiveness around the retrained model consistently.

Robustness. Most of the unlearning algorithms lack robustness. GNN-specific unlearning algorithms diverge in training due to unstable Hessian approximations or excessively large noise added to the parameters during the update. The general unlearning algorithms Kookmin and Seif degrade performance more for large numbers of unlearning requests than other unlearning algorithms, whereas Fanchuan is robust across scenarios with the exception of NBR sensitive item unlearning. The recommender-specific method SCIF is the most robust method with no divergences during unlearning and the lowest variance overall while being efficient, effective, and maintaining utility. An exception is NBR sensitive item unlearning for DNNTSP where all unlearning algorithms predict sensitive items for more users than the retrained model.

Influence of model architecture. For attention-based models, general methods such as Kookmin and Seif perform well for few unlearning requests but degrade as requests accumulate, while Fanchuan and SCIF largely preserve effectiveness and utility. For the recurrent model GRU4Rec, SCIF is the only method that consistently maintains effectiveness, whereas general methods show notable degradation and Fanchuan fails under sensitive item unlearning. For graph-based models, graph-specific methods are often efficient and highly effective but can be unstable under sequential unlearning, while SCIF provides the most robust overall performance, with general methods remaining competitive mainly for CF sensitive item unlearning.

A.4 Detailed Result Tables

Table 1: Results for the Food dataset (category: alcohol), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	Avg/Req
BPR	Retrain	0.43 ± 0.01	0.54 ± 0.01	0.80 ± 0.02	1.27 ± 0.03	0.88 ± 0.02	1.39 ± 0.03	34.20 ± 8.15	50.23 ± 7.05	147.02 ± 36.90
	Fanchuan	0.39 ± 0.01	0.50 ± 0.01	0.73 ± 0.03	1.16 ± 0.03	0.80 ± 0.03	1.26 ± 0.03	39.96 ± 5.36	56.57 ± 4.89	0.03 ± 0.00
	Kookmin	0.42 ± 0.01	0.54 ± 0.01	0.80 ± 0.01	1.26 ± 0.03	0.87 ± 0.01	1.38 ± 0.03	34.01 ± 8.00	50.17 ± 6.83	0.02 ± 0.00
	SCIF	0.43 ± 0.01	0.54 ± 0.01	0.80 ± 0.02	1.27 ± 0.03	0.88 ± 0.02	1.39 ± 0.03	34.20 ± 8.15	50.23 ± 7.05	0.04 ± 0.00
	Seif	0.00 ± 0.00	0.01 ± 0.00	0.01 ± 0.00	0.02 ± 0.01	0.01 ± 0.00	0.02 ± 0.00	33.46 ± 6.70	51.83 ± 8.18	0.02 ± 0.00
LightGCN	Retrain	0.70 ± 0.01	0.89 ± 0.01	1.31 ± 0.01	2.04 ± 0.02	1.41 ± 0.02	2.20 ± 0.02	37.27 ± 4.50	55.59 ± 5.89	346.69 ± 38.29
	CEU	1.16 ± 0.03	1.46 ± 0.03	2.17 ± 0.04	3.39 ± 0.05	2.31 ± 0.05	3.62 ± 0.06	33.27 ± 7.01	51.69 ± 4.75	0.05 ± 0.01
	Fanchuan	1.08 ± 0.01	1.35 ± 0.01	2.00 ± 0.02	3.09 ± 0.02	2.12 ± 0.01	3.28 ± 0.02	44.76 ± 4.71	60.01 ± 2.93	0.05 ± 0.00
	GIF	1.10 ± 0.02	1.42 ± 0.03	2.11 ± 0.06	3.36 ± 0.07	2.24 ± 0.05	3.58 ± 0.07	30.49 ± 2.90	48.58 ± 4.30	0.87 ± 0.18
	IDEA	0.54 ± 0.01	0.65 ± 0.01	0.93 ± 0.01	1.35 ± 0.01	0.94 ± 0.01	1.37 ± 0.02	31.12 ± 3.60	50.34 ± 3.76	0.04 ± 0.01
	Kookmin	1.15 ± 0.04	1.46 ± 0.04	2.17 ± 0.06	3.40 ± 0.09	2.31 ± 0.06	3.62 ± 0.08	32.94 ± 5.18	49.53 ± 5.93	0.03 ± 0.00
	SCIF	1.15 ± 0.04	1.46 ± 0.04	2.17 ± 0.06	3.40 ± 0.08	2.31 ± 0.06	3.62 ± 0.09	33.47 ± 5.07	50.06 ± 6.14	0.22 ± 0.01
SGL	Retrain	0.87 ± 0.01	1.08 ± 0.01	1.61 ± 0.01	2.45 ± 0.02	1.73 ± 0.01	2.64 ± 0.01	32.50 ± 2.99	49.79 ± 2.82	241.35 ± 21.70
	CEU	0.10 ± 0.05	0.14 ± 0.05	0.21 ± 0.09	0.37 ± 0.12	0.23 ± 0.09	0.40 ± 0.13	28.28 ± 19.86	46.09 ± 27.57	3.48 ± 0.87
	Fanchuan	1.72 ± 0.01	2.18 ± 0.01	3.22 ± 0.01	5.01 ± 0.02	3.38 ± 0.01	5.26 ± 0.02	33.51 ± 2.66	49.59 ± 3.73	0.41 ± 0.01
	GIF	0.10 ± 0.07	0.13 ± 0.08	0.20 ± 0.12	0.35 ± 0.18	0.21 ± 0.14	0.38 ± 0.19	14.78 ± 17.04	31.46 ± 28.48	3.27 ± 0.83
	Kookmin	1.46 ± 0.02	1.83 ± 0.03	2.72 ± 0.03	4.20 ± 0.06	2.88 ± 0.04	4.43 ± 0.06	29.54 ± 2.19	48.41 ± 1.93	0.13 ± 0.01
	SCIF	1.44 ± 0.03	1.81 ± 0.03	2.69 ± 0.04	4.15 ± 0.06	2.84 ± 0.04	4.39 ± 0.06	29.23 ± 3.12	46.19 ± 3.57	0.57 ± 0.12
	Seif	0.26 ± 0.03	0.36 ± 0.05	0.57 ± 0.09	0.95 ± 0.16	0.61 ± 0.10	1.02 ± 0.16	31.96 ± 11.54	53.11 ± 13.64	0.28 ± 0.03

Table 2: Results for the MovieLens dataset (category: health), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	Avg/Req
BPR	Retrain	8.06 ± 0.13	9.54 ± 0.13	7.33 ± 0.09	12.63 ± 0.11	35.19 ± 0.23	49.36 ± 0.17	0.03 ± 0.04	0.06 ± 0.07	110.69 ± 28.53
	Fanchuan	6.39 ± 0.10	7.90 ± 0.15	6.24 ± 0.15	11.08 ± 0.22	31.03 ± 0.63	45.52 ± 0.97	0.58 ± 0.28	1.50 ± 0.74	0.07 ± 0.00
	Kookmin	7.65 ± 1.62	8.85 ± 1.91	6.72 ± 1.50	11.25 ± 2.53	32.12 ± 6.08	44.51 ± 8.18	0.00 ± 0.00	0.00 ± 0.00	0.05 ± 0.01
	SCIF	7.34 ± 1.68	8.66 ± 2.00	6.66 ± 1.56	11.44 ± 2.70	32.03 ± 7.22	44.83 ± 10.03	0.38 ± 0.41	0.98 ± 0.99	0.12 ± 0.00
	Seif	0.16 ± 0.03	0.20 ± 0.03	0.10 ± 0.03	0.22 ± 0.03	1.34 ± 0.14	2.63 ± 0.14	0.00 ± 0.00	0.00 ± 0.00	0.06 ± 0.00
LightGCN	Retrain	7.78 ± 0.03	9.15 ± 0.03	6.78 ± 0.03	11.94 ± 0.05	33.78 ± 0.12	47.68 ± 0.16	0.01 ± 0.02	0.03 ± 0.03	1344.00 ± 80.98
	CEU	8.91 ± 0.05	10.50 ± 0.04	7.93 ± 0.06	13.76 ± 0.07	37.48 ± 0.17	52.08 ± 0.20	0.47 ± 0.28	1.23 ± 0.47	0.17 ± 0.03
	Fanchuan	7.48 ± 0.16	8.94 ± 0.22	6.76 ± 0.16	11.85 ± 0.35	33.76 ± 0.51	48.05 ± 0.74	4.60 ± 0.34	5.72 ± 0.46	0.87 ± 0.26
	GIF	7.30 ± 0.50	9.07 ± 0.45	7.13 ± 0.32	12.75 ± 0.34	34.75 ± 0.92	49.84 ± 0.75	0.08 ± 0.13	0.21 ± 0.25	2.89 ± 0.78
	Kookmin	8.89 ± 0.15	10.47 ± 0.17	7.95 ± 0.17	13.72 ± 0.25	37.28 ± 0.34	51.65 ± 0.37	0.11 ± 0.07	0.44 ± 0.25	0.92 ± 0.30
	SCIF	8.99 ± 0.05	10.58 ± 0.04	8.00 ± 0.05	13.84 ± 0.06	37.68 ± 0.15	52.20 ± 0.19	0.45 ± 0.23	1.21 ± 0.31	0.18 ± 0.00
	Seif	0.76 ± 0.39	0.94 ± 0.48	0.61 ± 0.28	1.21 ± 0.55	4.58 ± 2.20	8.49 ± 4.00	0.00 ± 0.00	0.03 ± 0.07	5.53 ± 5.75
SGL	Retrain	6.93 ± 0.04	8.31 ± 0.03	6.18 ± 0.04	11.02 ± 0.02	31.86 ± 0.13	45.70 ± 0.12	0.00 ± 0.00	0.00 ± 0.00	1370.81 ± 32.45
	CEU	5.58 ± 0.11	6.35 ± 0.15	4.69 ± 0.14	7.76 ± 0.25	25.12 ± 0.18	35.46 ± 0.15	0.00 ± 0.00	0.00 ± 0.00	0.52 ± 0.11
	Kookmin	9.09 ± 0.05	10.73 ± 0.07	8.17 ± 0.09	13.96 ± 0.14	39.36 ± 0.14	53.96 ± 0.23	0.34 ± 0.10	0.70 ± 0.17	2.99 ± 0.85
	SCIF	7.97 ± 0.07	9.56 ± 0.08	7.37 ± 0.08	12.83 ± 0.11	35.54 ± 0.18	49.79 ± 0.28	0.00 ± 0.00	0.02 ± 0.03	5.20 ± 2.48
	Seif	0.10 ± 0.06	0.16 ± 0.09	0.04 ± 0.02	0.14 ± 0.08	0.80 ± 0.69	2.30 ± 1.58	0.00 ± 0.00	0.00 ± 0.00	5.04 ± 1.29

Table 3: Results for the Instacart dataset (category: meat), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	
DNNTSP	Retrain	29.17 $\pm$ 0.21	30.38 $\pm$ 0.16	24.89 $\pm$ 0.08	32.64 $\pm$ 0.07	77.23 $\pm$ 0.11	83.36 $\pm$ 0.04	12.24 $\pm$ 1.70	37.16 $\pm$ 4.59	1442.49 $\pm$ 5.23
	Fanchuan	<u>18.43</u> $\pm$ 4.92	<u>20.27</u> $\pm$ 5.44	17.56 $\pm$ 4.39	24.29 $\pm$ 6.32	62.80 $\pm$ 12.49	71.22 $\pm$ 12.57	43.37 $\pm$ 22.53	<u>74.19</u> $\pm$ 14.35	0.91 $\pm$ 0.01
	Kookmin	16.74 $\pm$ 2.02	19.96 $\pm$ 1.79	<u>19.66</u> $\pm$ 1.51	<u>29.05</u> $\pm$ 1.60	<u>70.94</u> $\pm$ 2.62	<u>80.64</u> $\pm$ 2.00	35.45 $\pm$ 9.52	74.56 $\pm$ 4.61	<u>0.03</u> $\pm$ 0.00
	SCIF	26.97 $\pm$ 2.33	28.55 $\pm$ 1.92	23.74 $\pm$ 1.28	31.74 $\pm$ 0.98	75.93 $\pm$ 1.77	82.87 $\pm$ 0.56	53.90 $\pm$ 4.82	82.17 $\pm$ 2.06	0.02 $\pm$ 0.00
	Seif	0.21 $\pm$ 0.11	0.24 $\pm$ 0.10	0.17 $\pm$ 0.08	0.29 $\pm$ 0.10	1.31 $\pm$ 0.64	2.10 $\pm$ 0.68	<u>40.05</u> $\pm$ 54.55	59.88 $\pm$ 54.51	0.03 $\pm$ 0.00
Sets2Sets	Retrain	16.89 $\pm$ 8.72	15.78 $\pm$ 8.11	11.90 $\pm$ 6.01	13.35 $\pm$ 6.73	53.04 $\pm$ 24.30	56.21 $\pm$ 24.72	20.85 $\pm$ 44.28	62.56 $\pm$ 23.03	311.44 $\pm$ 65.71
	Fanchuan	6.50 $\pm$ 1.89	5.96 $\pm$ 1.62	4.08 $\pm$ 1.17	4.62 $\pm$ 1.03	29.45 $\pm$ 5.31	32.63 $\pm$ 4.88	55.40 $\pm$ 48.31	66.40 $\pm$ 46.86	0.07 $\pm$ 0.00
	Kookmin	21.04 $\pm$ 0.76	19.32 $\pm$ 0.52	14.58 $\pm$ 0.28	15.75 $\pm$ 0.00	64.37 $\pm$ 0.47	66.51 $\pm$ 0.03	28.45 $\pm$ 0.80	34.73 $\pm$ 1.08	0.01 $\pm$ 0.00
	SCIF	<u>18.50</u> $\pm$ 4.06	<u>16.95</u> $\pm$ 3.61	<u>14.09</u> $\pm$ 0.78	<u>15.12</u> $\pm$ 0.74	<u>62.70</u> $\pm$ 1.28	<u>64.76</u> $\pm$ 1.02	30.05 $\pm$ 4.18	<u>58.87</u> $\pm$ 31.43	0.02 $\pm$ 0.00
	Seif	8.02 $\pm$ 9.17	7.54 $\pm$ 8.52	5.66 $\pm$ 6.16	6.49 $\pm$ 6.93	29.45 $\pm$ 27.16	32.69 $\pm$ 28.12	70.75 $\pm$ 40.09	80.04 $\pm$ 29.32	<u>0.01</u> $\pm$ 0.00

Table 4: Results for the Dunnhumby dataset (category: alcohol), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	
DNNTSP	Retrain	7.11 $\pm$ 0.31	7.12 $\pm$ 0.37	7.25 $\pm$ 0.24	8.98 $\pm$ 0.42	30.42 $\pm$ 0.78	36.39 $\pm$ 1.11	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	95.53 $\pm$ 4.08
	Fanchuan	2.59 $\pm$ 1.13	3.00 $\pm$ 1.12	2.68 $\pm$ 1.02	4.25 $\pm$ 1.27	14.47 $\pm$ 7.21	21.43 $\pm$ 8.55	0.57 $\pm$ 0.26	1.21 $\pm$ 0.29	0.49 $\pm$ 0.01
	Kookmin	<u>5.93</u> $\pm$ 0.52	6.19 $\pm$ 0.52	6.94 $\pm$ 0.17	8.78 $\pm$ 0.38	29.29 $\pm$ 0.59	35.95 $\pm$ 1.22	0.21 $\pm$ 0.23	17.62 $\pm$ 38.90	<u>0.02</u> $\pm$ 0.00
	SCIF	6.95 $\pm$ 0.22	7.00 $\pm$ 0.32	7.16 $\pm$ 0.15	8.84 $\pm$ 0.35	29.99 $\pm$ 0.65	36.05 $\pm$ 1.29	<u>0.11</u> $\pm$ 0.16	<u>0.14</u> $\pm$ 0.15	0.01 $\pm$ 0.00
	Seif	0.11 $\pm$ 0.06	0.12 $\pm$ 0.06	0.12 $\pm$ 0.08	0.18 $\pm$ 0.09	0.72 $\pm$ 0.43	1.16 $\pm$ 0.44	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.02 $\pm$ 0.00
Sets2Sets	Retrain	9.88 $\pm$ 0.61	9.74 $\pm$ 0.55	8.49 $\pm$ 0.89	10.74 $\pm$ 0.98	35.44 $\pm$ 2.46	41.46 $\pm$ 2.25	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	101.48 $\pm$ 1.54
	Fanchuan	2.38 $\pm$ 1.22	2.33 $\pm$ 1.14	2.12 $\pm$ 1.08	2.44 $\pm$ 1.10	7.98 $\pm$ 3.93	9.09 $\pm$ 4.15	0.18 $\pm$ 0.13	0.25 $\pm$ 0.20	0.12 $\pm$ 0.00
	Kookmin	9.30 $\pm$ 0.21	9.38 $\pm$ 0.19	8.27 $\pm$ 0.53	11.01 $\pm$ 0.40	35.73 $\pm$ 0.43	42.70 $\pm$ 0.52	<u>0.07</u> $\pm$ 0.10	<u>0.18</u> $\pm$ 0.13	0.02 $\pm$ 0.00
	SCIF	3.37 $\pm$ 2.39	4.41 $\pm$ 1.80	3.74 $\pm$ 2.17	<u>7.11</u> $\pm$ 1.71	<u>17.59</u> $\pm$ 10.46	<u>29.87</u> $\pm$ 7.04	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.02 $\pm$ 0.00
	Seif	<u>4.26</u> $\pm$ 0.12	<u>4.83</u> $\pm$ 0.13	<u>4.43</u> $\pm$ 0.20	6.61 $\pm$ 0.30	17.51 $\pm$ 0.80	25.08 $\pm$ 0.76	4.23 $\pm$ 0.93	6.32 $\pm$ 0.99	<u>0.02</u> $\pm$ 0.00

Table 5: Results for the Books dataset (category: health), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	
GRU4Rec	Retrain	2.53 $\pm$ 0.08	3.07 $\pm$ 0.10	4.77 $\pm$ 0.15	6.90 $\pm$ 0.22	4.77 $\pm$ 0.15	6.90 $\pm$ 0.22	58.60 $\pm$ 2.69	77.62 $\pm$ 1.71	428.05 $\pm$ 55.45
	Fanchuan	0.37 $\pm$ 0.06	0.48 $\pm$ 0.07	0.74 $\pm$ 0.11	1.19 $\pm$ 0.15	0.74 $\pm$ 0.11	1.19 $\pm$ 0.15	72.78 $\pm$ 17.27	92.49 $\pm$ 4.47	0.13 $\pm$ 0.00
	Kookmin	0.77 $\pm$ 0.11	1.04 $\pm$ 0.15	<u>1.58</u> $\pm$ 0.23	<u>2.66</u> $\pm$ 0.35	<u>1.58</u> $\pm$ 0.23	<u>2.66</u> $\pm$ 0.35	70.75 $\pm$ 7.79	89.79 $\pm$ 4.08	0.12 $\pm$ 0.00
	SCIF	<u>0.69</u> $\pm$ 0.15	<u>0.99</u> $\pm$ 0.20	1.61 $\pm$ 0.30	2.81 $\pm$ 0.49	1.61 $\pm$ 0.30	2.81 $\pm$ 0.49	<u>63.19</u> $\pm$ 23.74	<u>82.90</u> $\pm$ 12.70	0.15 $\pm$ 0.00
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	59.02 $\pm$ 35.49	79.33 $\pm$ 29.60	<u>0.12</u> $\pm$ 0.00
NARM	Retrain	2.34 $\pm$ 0.02	2.85 $\pm$ 0.02	4.47 $\pm$ 0.04	6.53 $\pm$ 0.06	4.47 $\pm$ 0.04	6.53 $\pm$ 0.06	56.70 $\pm$ 3.42	77.30 $\pm$ 1.25	350.34 $\pm$ 97.21
	Fanchuan	0.51 $\pm$ 0.03	0.65 $\pm$ 0.04	1.01 $\pm$ 0.06	1.59 $\pm$ 0.09	1.01 $\pm$ 0.06	1.59 $\pm$ 0.09	<u>63.46</u> $\pm$ 21.58	<u>82.89</u> $\pm$ 14.81	0.13 $\pm$ 0.00
	Kookmin	<u>0.51</u> $\pm$ 0.08	<u>0.70</u> $\pm$ 0.10	<u>1.06</u> $\pm$ 0.17	<u>1.83</u> $\pm$ 0.29	<u>1.06</u> $\pm$ 0.17	<u>1.83</u> $\pm$ 0.29	79.93 $\pm$ 9.23	94.55 $\pm$ 3.37	<u>0.12</u> $\pm$ 0.00
	SCIF	0.55 $\pm$ 0.15	0.85 $\pm$ 0.16	1.32 $\pm$ 0.28	2.51 $\pm$ 0.34	1.32 $\pm$ 0.28	2.51 $\pm$ 0.34	66.64 $\pm$ 18.71	83.02 $\pm$ 9.73	0.15 $\pm$ 0.00
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.01	0.00 $\pm$ 0.00	0.00 $\pm$ 0.01	37.14 $\pm$ 19.02	68.34 $\pm$ 15.77	0.12 $\pm$ 0.00
SASRec	Retrain	1.62 $\pm$ 0.01	2.04 $\pm$ 0.03	3.18 $\pm$ 0.04	4.84 $\pm$ 0.08	3.18 $\pm$ 0.04	4.84 $\pm$ 0.08	59.27 $\pm$ 3.37	77.31 $\pm$ 1.96	786.96 $\pm$ 448.36
	Fanchuan	0.28 $\pm$ 0.03	0.36 $\pm$ 0.04	0.55 $\pm$ 0.06	0.88 $\pm$ 0.10	0.55 $\pm$ 0.06	0.88 $\pm$ 0.10	45.77 $\pm$ 22.73	74.98 $\pm$ 22.54	0.15 $\pm$ 0.00
	Kookmin	0.03 $\pm$ 0.07	0.04 $\pm$ 0.09	0.07 $\pm$ 0.14	0.12 $\pm$ 0.24	0.07 $\pm$ 0.14	0.12 $\pm$ 0.24	<u>41.79</u> $\pm$ 53.26	60.00 $\pm$ 54.77	<u>0.12</u> $\pm$ 0.00
	SCIF	<u>0.04</u> $\pm$ 0.01	<u>0.08</u> $\pm$ 0.03	<u>0.09</u> $\pm$ 0.03	<u>0.28</u> $\pm$ 0.08	<u>0.09</u> $\pm$ 0.03	<u>0.28</u> $\pm$ 0.08	56.97 $\pm$ 11.98	85.20 $\pm$ 6.24	0.13 $\pm$ 0.00
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	40.00 $\pm$ 54.77	80.00 $\pm$ 44.72	0.12 $\pm$ 0.00
SRGNN	Retrain	3.56 $\pm$ 0.02	4.12 $\pm$ 0.02	6.06 $\pm$ 0.02	8.29 $\pm$ 0.04	6.06 $\pm$ 0.02	8.29 $\pm$ 0.04	56.49 $\pm$ 1.87	78.02 $\pm$ 0.93	763.09 $\pm$ 73.15
	CEU	0.01 $\pm$ 0.01	0.01 $\pm$ 0.01	0.01 $\pm$ 0.01	0.03 $\pm$ 0.02	0.01 $\pm$ 0.01	0.03 $\pm$ 0.02	83.50 $\pm$ 36.77	99.87 $\pm$ 0.21	0.18 $\pm$ 0.01
	Fanchuan	<u>0.12</u> $\pm$ 0.02	0.17 $\pm$ 0.03	0.26 $\pm$ 0.06	0.45 $\pm$ 0.11	0.26 $\pm$ 0.06	0.45 $\pm$ 0.11	61.68 $\pm$ 27.69	97.33 $\pm$ 3.40	0.27 $\pm$ 0.00
	IDEA	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.01	0.00 $\pm$ 0.00	0.01 $\pm$ 0.01	42.30 $\pm$ 3.13	63.96 $\pm$ 0.61	0.12 $\pm$ 0.00
	Kookmin	1.04 $\pm$ 0.13	1.25 $\pm$ 0.15	1.90 $\pm$ 0.22	2.76 $\pm$ 0.33	1.90 $\pm$ 0.22	2.76 $\pm$ 0.33	<u>50.42</u> $\pm$ 6.42	<u>72.86</u> $\pm$ 6.03	<u>0.13</u> $\pm$ 0.01
	SCIF	0.11 $\pm$ 0.04	<u>0.22</u> $\pm$ 0.06	<u>0.29</u> $\pm$ 0.10	<u>0.72</u> $\pm$ 0.22	<u>0.29</u> $\pm$ 0.10	<u>0.72</u> $\pm$ 0.22	58.65 $\pm$ 23.70	83.21 $\pm$ 10.02	0.19 $\pm$ 0.01
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	<u>64.26</u> $\pm$ 49.02	80.29 $\pm$ 43.65	0.14 $\pm$ 0.00

Table 6: Results for the 30Music dataset (category: explicit), averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Sens. Items (%)		Time (min)
		@10	@20	@10	@20	@10	@20	@10	@20	Avg/Req
GRU4Rec	Retrain	14.57 $\pm$ 0.28	15.29 $\pm$ 0.28	19.98 $\pm$ 0.35	22.83 $\pm$ 0.39	19.98 $\pm$ 0.35	22.83 $\pm$ 0.39	1.84 $\pm$ 1.18	4.04 $\pm$ 0.96	932.98 $\pm$ 266.80
	Fanchuan	<u>1.13</u> $\pm$ 1.10	<u>1.29</u> $\pm$ 1.22	<u>1.94</u> $\pm$ 1.84	<u>2.59</u> $\pm$ 2.32	<u>1.94</u> $\pm$ 1.84	<u>2.59</u> $\pm$ 2.32	5.33 $\pm$ 4.09	8.27 $\pm$ 4.71	1.51 $\pm$ 0.20
	Kookmin	0.50 $\pm$ 0.12	0.60 $\pm$ 0.13	0.92 $\pm$ 0.21	1.28 $\pm$ 0.27	0.92 $\pm$ 0.21	1.28 $\pm$ 0.27	<u>9.10</u> $\pm$ 3.53	<u>12.40</u> $\pm$ 3.10	<u>1.07</u> $\pm$ 0.40
	SCIF	14.59 $\pm$ 0.39	15.32 $\pm$ 0.40	20.04 $\pm$ 0.48	22.90 $\pm$ 0.54	20.04 $\pm$ 0.48	22.90 $\pm$ 0.54	<u>9.10</u> $\pm$ 3.53	<u>12.40</u> $\pm$ 3.10	0.72 $\pm$ 0.08
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	<u>9.10</u> $\pm$ 3.53	<u>12.40</u> $\pm$ 3.10	1.14 $\pm$ 0.13
NARM	Retrain	12.81 $\pm$ 0.33	13.62 $\pm$ 0.31	19.05 $\pm$ 0.32	22.21 $\pm$ 0.27	19.05 $\pm$ 0.32	22.21 $\pm$ 0.27	2.66 $\pm$ 1.27	5.28 $\pm$ 1.46	909.20 $\pm$ 214.21
	Fanchuan	<u>1.09</u> $\pm$ 1.25	<u>1.24</u> $\pm$ 1.40	<u>1.94</u> $\pm$ 2.25	<u>2.54</u> $\pm$ 2.85	<u>1.94</u> $\pm$ 2.25	<u>2.54</u> $\pm$ 2.85	<u>7.05</u> $\pm$ 12.43	8.56 $\pm$ 15.04	1.47 $\pm$ 0.15
	Kookmin	0.34 $\pm$ 0.11	0.43 $\pm$ 0.12	0.68 $\pm$ 0.21	1.01 $\pm$ 0.28	0.68 $\pm$ 0.21	1.01 $\pm$ 0.28	7.53 $\pm$ 2.22	9.46 $\pm$ 2.19	1.28 $\pm$ 0.02
	SCIF	12.65 $\pm$ 0.27	13.45 $\pm$ 0.25	18.87 $\pm$ 0.18	22.05 $\pm$ 0.17	18.87 $\pm$ 0.18	22.05 $\pm$ 0.17	7.04 $\pm$ 1.93	<u>9.11</u> $\pm$ 2.50	0.70 $\pm$ 0.05
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	7.53 $\pm$ 2.22	9.46 $\pm$ 2.19	<u>1.21</u> $\pm$ 0.11
SASRec	Retrain	8.15 $\pm$ 0.09	9.02 $\pm$ 0.09	13.98 $\pm$ 0.12	17.41 $\pm$ 0.13	13.98 $\pm$ 0.12	17.41 $\pm$ 0.13	1.97 $\pm$ 0.73	4.60 $\pm$ 0.59	810.28 $\pm$ 0.26
	Fanchuan	<u>2.20</u> $\pm$ 0.32	<u>2.56</u> $\pm$ 0.36	<u>4.18</u> $\pm$ 0.59	<u>5.58</u> $\pm$ 0.77	<u>4.18</u> $\pm$ 0.59	<u>5.58</u> $\pm$ 0.77	6.34 $\pm$ 3.46	7.95 $\pm$ 3.74	1.80 $\pm$ 0.21
	Kookmin	0.06 $\pm$ 0.05	0.09 $\pm$ 0.07	0.13 $\pm$ 0.11	0.23 $\pm$ 0.18	0.13 $\pm$ 0.11	0.23 $\pm$ 0.18	<u>7.43</u> $\pm$ 0.75	<u>9.81</u> $\pm$ 0.46	0.70 $\pm$ 0.50
	SCIF	7.48 $\pm$ 0.61	8.40 $\pm$ 0.55	13.53 $\pm$ 0.37	17.18 $\pm$ 0.16	13.53 $\pm$ 0.37	17.18 $\pm$ 0.16	7.90 $\pm$ 2.97	<u>10.43</u> $\pm$ 3.62	<u>0.83</u> $\pm$ 0.08
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.01	0.00 $\pm$ 0.00	0.00 $\pm$ 0.01	8.73 $\pm$ 2.67	11.44 $\pm$ 3.28	<u>1.21</u> $\pm$ 0.07
SRGNN	Retrain	16.30 $\pm$ 0.12	16.87 $\pm$ 0.12	20.78 $\pm$ 0.08	23.06 $\pm$ 0.08	20.78 $\pm$ 0.08	23.06 $\pm$ 0.08	0.74 $\pm$ 0.85	3.22 $\pm$ 0.96	1088.07 $\pm$ 258.96
	CEU	7.63 $\pm$ 5.94	7.96 $\pm$ 6.05	9.88 $\pm$ 7.04	11.21 $\pm$ 7.54	9.88 $\pm$ 7.04	11.21 $\pm$ 7.54	3.91 $\pm$ 3.63	5.47 $\pm$ 5.15	0.19 $\pm$ 0.04
	Fanchuan	5.27 $\pm$ 0.55	5.53 $\pm$ 0.56	6.97 $\pm$ 0.65	8.02 $\pm$ 0.68	6.97 $\pm$ 0.65	8.02 $\pm$ 0.68	<u>4.60</u> $\pm$ 5.49	<u>6.76</u> $\pm$ 7.01	4.31 $\pm$ 0.43
	Kookmin	<u>11.35</u> $\pm$ 0.64	<u>11.82</u> $\pm$ 0.67	<u>14.60</u> $\pm$ 0.89	<u>16.46</u> $\pm$ 1.05	<u>14.60</u> $\pm$ 0.89	<u>16.46</u> $\pm$ 1.05	11.67 $\pm$ 4.65	16.63 $\pm$ 5.89	1.21 $\pm$ 0.03
	SCIF	16.00 $\pm$ 0.47	16.60 $\pm$ 0.46	20.69 $\pm$ 0.25	23.01 $\pm$ 0.19	20.69 $\pm$ 0.25	23.01 $\pm$ 0.19	8.55 $\pm$ 3.04	12.27 $\pm$ 3.41	<u>1.00</u> $\pm$ 0.05
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	8.73 $\pm$ 3.08	12.77 $\pm$ 3.81	2.53 $\pm$ 0.06
	Seif	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	8.73 $\pm$ 3.08	12.77 $\pm$ 3.81	2.53 $\pm$ 0.06

Table 7: Results for the NowP dataset, averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Time (min)
		@10	@20	@10	@20	@10	@20	Avg/Req
GRU4Rec	Original	76.88 $\pm$ 0.16	77.07 $\pm$ 0.16	79.34 $\pm$ 0.13	80.10 $\pm$ 0.13	79.34 $\pm$ 0.13	80.10 $\pm$ 0.13	626.88 $\pm$ 804.7
	Retrain	79.69 $\pm$ 0.46	79.93 $\pm$ 0.45	82.48 $\pm$ 0.40	83.44 $\pm$ 0.38	82.48 $\pm$ 0.40	83.44 $\pm$ 0.38	906.05 $\pm$ 3.17
	Fanchuan	<u>74.05</u> $\pm$ 1.08	<u>74.20</u> $\pm$ 1.09	<u>76.05</u> $\pm$ 1.15	<u>76.63</u> $\pm$ 1.17	<u>76.05</u> $\pm$ 1.15	<u>76.63</u> $\pm$ 1.17	<u>24.12</u> $\pm$ 2.36
	Kookmin	66.42 $\pm$ 0.35	66.60 $\pm$ 0.34	68.40 $\pm$ 0.34	69.16 $\pm$ 0.34	68.40 $\pm$ 0.34	69.16 $\pm$ 0.34	28.39 $\pm$ 2.15
	SCIF	77.49 $\pm$ 0.13	77.69 $\pm$ 0.12	79.98 $\pm$ 0.09	80.74 $\pm$ 0.09	79.98 $\pm$ 0.09	80.74 $\pm$ 0.09	214.97 $\pm$ 50.66
	Seif	64.04 $\pm$ 1.07	64.24 $\pm$ 1.07	66.14 $\pm$ 1.06	66.92 $\pm$ 1.06	66.14 $\pm$ 1.06	66.92 $\pm$ 1.06	11.14 $\pm$ 0.09
NARM	Original	76.78 $\pm$ 0.54	76.97 $\pm$ 0.54	79.50 $\pm$ 0.33	80.25 $\pm$ 0.30	79.50 $\pm$ 0.33	80.25 $\pm$ 0.30	408.68 $\pm$ 451.88
	Retrain	81.14 $\pm$ 0.36	81.35 $\pm$ 0.35	83.83 $\pm$ 0.16	84.67 $\pm$ 0.14	83.83 $\pm$ 0.16	84.67 $\pm$ 0.14	1287.81 $\pm$ 229.73
	Fanchuan	<u>67.28</u> $\pm$ 8.43	<u>67.53</u> $\pm$ 8.30	<u>70.67</u> $\pm$ 6.59	<u>71.68</u> $\pm$ 6.06	<u>70.67</u> $\pm$ 6.59	<u>71.68</u> $\pm$ 6.06	26.15 $\pm$ 2.3
	Kookmin	52.84 $\pm$ 1.53	53.11 $\pm$ 1.53	55.16 $\pm$ 1.62	56.20 $\pm$ 1.63	55.16 $\pm$ 1.62	56.20 $\pm$ 1.63	<u>15.53</u> $\pm$ 1.69
	SCIF	77.39 $\pm$ 0.59	77.58 $\pm$ 0.58	80.14 $\pm$ 0.36	80.89 $\pm$ 0.33	80.14 $\pm$ 0.36	80.89 $\pm$ 0.33	128.87 $\pm$ 14.4
	Seif	62.15 $\pm$ 2.63	62.37 $\pm$ 2.60	64.72 $\pm$ 2.49	65.59 $\pm$ 2.38	64.72 $\pm$ 2.49	65.59 $\pm$ 2.38	11.09 $\pm$ 0.06
SASRec	Original	49.24 $\pm$ 0.51	49.85 $\pm$ 0.51	64.43 $\pm$ 0.18	66.82 $\pm$ 0.16	64.43 $\pm$ 0.18	66.82 $\pm$ 0.16	292.81 $\pm$ 193.97
	Retrain	58.56 $\pm$ 2.08	59.13 $\pm$ 2.03	71.20 $\pm$ 0.49	73.41 $\pm$ 0.33	71.20 $\pm$ 0.49	73.41 $\pm$ 0.33	1224.19 $\pm$ 180.37
	Fanchuan	<u>43.45</u> $\pm$ 0.45	<u>44.07</u> $\pm$ 0.43	<u>58.38</u> $\pm$ 0.75	<u>60.81</u> $\pm$ 0.72	<u>58.38</u> $\pm$ 0.75	<u>60.81</u> $\pm$ 0.72	38.25 $\pm$ 3.9
	Kookmin	25.25 $\pm$ 2.79	25.75 $\pm$ 2.78	32.69 $\pm$ 2.94	34.65 $\pm$ 2.93	32.69 $\pm$ 2.94	34.65 $\pm$ 2.93	<u>16.96</u> $\pm$ 1.59
	SCIF	49.64 $\pm$ 0.56	50.25 $\pm$ 0.55	64.95 $\pm$ 0.19	67.35 $\pm$ 0.17	64.95 $\pm$ 0.19	67.35 $\pm$ 0.17	63.85 $\pm$ 6.62
	Seif	0.11 $\pm$ 0.09	0.17 $\pm$ 0.12	0.25 $\pm$ 0.19	0.49 $\pm$ 0.30	0.25 $\pm$ 0.19	0.49 $\pm$ 0.30	11.08 $\pm$ 0.06
SRGNN	Original	70.20 $\pm$ 0.18	70.45 $\pm$ 0.18	72.85 $\pm$ 0.13	73.85 $\pm$ 0.11	72.85 $\pm$ 0.13	73.85 $\pm$ 0.11	355.87 $\pm$ 350.49
	Retrain	74.75 $\pm$ 0.05	75.04 $\pm$ 0.05	77.52 $\pm$ 0.05	78.63 $\pm$ 0.05	77.52 $\pm$ 0.05	78.63 $\pm$ 0.05	1046.12 $\pm$ 26.54
	CEU	<u>70.76</u> $\pm$ 0.16	71.01 $\pm$ 0.16	73.43 $\pm$ 0.10	<u>74.44</u> $\pm$ 0.08	73.43 $\pm$ 0.10	<u>74.44</u> $\pm$ 0.08	1.96 $\pm$ 0.35
	Fanchuan	69.53 $\pm$ 0.84	<u>69.72</u> $\pm$ 0.84	<u>71.59</u> $\pm$ 0.84	72.35 $\pm$ 0.84	<u>71.59</u> $\pm$ 0.84	72.35 $\pm$ 0.84	105.59 $\pm$ 9.3
	Kookmin	48.47 $\pm$ 2.14	48.63 $\pm$ 2.13	49.80 $\pm$ 2.20	50.45 $\pm$ 2.30	49.80 $\pm$ 2.20	50.45 $\pm$ 2.30	<u>22.06</u> $\pm$ 2.86
	SCIF	70.76 $\pm$ 0.16	71.01 $\pm$ 0.16	73.43 $\pm$ 0.10	74.45 $\pm$ 0.08	73.43 $\pm$ 0.10	74.45 $\pm$ 0.08	56.79 $\pm$ 5.88
	Seif	64.85 $\pm$ 4.54	65.12 $\pm$ 4.50	67.31 $\pm$ 4.21	68.40 $\pm$ 4.06	67.31 $\pm$ 4.21	68.40 $\pm$ 4.06	<u>11.3</u> $\pm$ 0.44

Table 8: Results for the RSC15 dataset, averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Time (min)
		@10	@20	@10	@20	@10	@20	Avg/Req
GRU4Rec	Original	37.65 ± 0.03	39.91 ± 0.03	58.84 ± 0.04	67.74 ± 0.05	58.84 ± 0.04	67.74 ± 0.05	413.87 ± 118.81
	Retrain	38.46 ± 0.07	40.77 ± 0.06	60.10 ± 0.05	69.19 ± 0.04	60.10 ± 0.05	69.19 ± 0.04	364 ± 42.05
	Fanchuan	<u>37.12</u> ± 0.14	<u>39.45</u> ± 0.15	<u>58.48</u> ± 0.08	<u>67.67</u> ± 0.11	<u>58.48</u> ± 0.08	<u>67.67</u> ± 0.11	1.28 ± 0.09
	Kookmin	14.93 ± 4.30	15.86 ± 4.52	22.97 ± 6.50	26.61 ± 7.35	22.97 ± 6.50	26.61 ± 7.35	<u>1.16</u> ± 0.02
	SCIF	38.44 ± 0.04	40.74 ± 0.03	60.07 ± 0.05	69.15 ± 0.05	60.07 ± 0.05	69.15 ± 0.05	1.82 ± 0.07
	Seif	0.01 ± 0.01	0.01 ± 0.02	0.02 ± 0.02	0.04 ± 0.04	0.02 ± 0.02	0.04 ± 0.04	1.15 ± 0
NARM	Original	37.25 ± 0.02	39.52 ± 0.02	58.45 ± 0.02	67.39 ± 0.02	58.45 ± 0.02	67.39 ± 0.02	539.19 ± 77.34
	Retrain	37.99 ± 0.04	40.31 ± 0.04	59.62 ± 0.06	68.75 ± 0.06	59.62 ± 0.06	68.75 ± 0.06	765.09 ± 106.15
	Fanchuan	<u>35.94</u> ± 0.14	<u>38.30</u> ± 0.13	<u>57.00</u> ± 0.15	<u>66.28</u> ± 0.10	<u>57.00</u> ± 0.15	<u>66.28</u> ± 0.10	1.14 ± 0.01
	Kookmin	10.21 ± 0.39	10.85 ± 0.50	15.87 ± 0.62	18.38 ± 1.06	15.87 ± 0.62	18.38 ± 1.06	1.05 ± 0.01
	SCIF	38.03 ± 0.03	40.34 ± 0.02	59.66 ± 0.03	68.79 ± 0.02	59.66 ± 0.03	68.79 ± 0.02	1.46 ± 0
	Seif	0.00 ± 0.01	0.01 ± 0.01	0.01 ± 0.02	0.02 ± 0.03	0.01 ± 0.02	0.02 ± 0.03	<u>1.06</u> ± 0.01
SASRec	Original	32.42 ± 0.05	34.82 ± 0.04	52.31 ± 0.06	61.79 ± 0.06	52.31 ± 0.06	61.79 ± 0.06	561.27 ± 57.35
	Retrain	33.12 ± 0.05	35.59 ± 0.06	53.41 ± 0.03	63.11 ± 0.08	53.41 ± 0.03	63.11 ± 0.08	843.67 ± 195.32
	Fanchuan	<u>31.25</u> ± 0.39	<u>33.69</u> ± 0.40	<u>50.88</u> ± 0.58	<u>60.51</u> ± 0.61	<u>50.88</u> ± 0.58	<u>60.51</u> ± 0.61	1.31 ± 0.08
	Kookmin	2.10 ± 0.46	2.84 ± 0.57	4.12 ± 0.94	7.07 ± 1.42	4.12 ± 0.94	7.07 ± 1.42	1.08 ± 0.1
	SCIF	33.09 ± 0.05	35.55 ± 0.05	53.40 ± 0.06	63.07 ± 0.06	53.40 ± 0.06	63.07 ± 0.06	1.27 ± 0.1
	Seif	0.01 ± 0.01	0.01 ± 0.01	0.02 ± 0.03	0.03 ± 0.04	0.02 ± 0.03	0.03 ± 0.04	<u>1.25</u> ± 0.08
SRGNN	Original	39.23 ± 0.01	41.41 ± 0.01	60.18 ± 0.01	68.72 ± 0.01	60.18 ± 0.01	68.72 ± 0.01	660.73 ± 154.37
	Retrain	40.04 ± 0.04	42.27 ± 0.04	61.43 ± 0.03	70.16 ± 0.02	61.43 ± 0.03	70.16 ± 0.02	1441.69 ± 6.4
	CEU	11.46 ± 9.32	13.30 ± 10.55	21.02 ± 16.66	28.28 ± 21.61	21.02 ± 16.66	28.28 ± 21.61	1.81 ± 0.02
	Fanchuan	30.37 ± 0.21	32.76 ± 0.20	49.95 ± 0.26	59.33 ± 0.23	49.95 ± 0.26	59.33 ± 0.23	1.51 ± 0.08
	Kookmin	<u>31.55</u> ± 2.18	<u>33.94</u> ± 2.20	<u>51.14</u> ± 3.20	<u>60.55</u> ± 3.30	<u>51.14</u> ± 3.20	<u>60.55</u> ± 3.30	1.13 ± 0.09
	SCIF	40.05 ± 0.01	42.27 ± 0.01	61.43 ± 0.01	70.15 ± 0.01	61.43 ± 0.01	70.15 ± 0.01	1.24 ± 0.11
	Seif	0.03 ± 0.02	0.04 ± 0.02	0.07 ± 0.04	0.10 ± 0.05	0.07 ± 0.04	0.10 ± 0.05	<u>1.2</u> ± 0.01

Table 9: Results for the Dunnhumby dataset, averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Time (min)
		@10	@20	@10	@20	@10	@20	Avg/Req
DNNTSP	Original	6.79 ± 0.25	6.79 ± 0.35	6.90 ± 0.17	8.52 ± 0.45	29.09 ± 0.75	34.75 ± 1.61	102.18 ± 5.86
	Retrain	6.65 ± 0.65	6.70 ± 0.67	6.98 ± 0.17	8.59 ± 0.46	29.46 ± 0.56	35.40 ± 1.51	100.22 ± 2.39
	CEU	6.55 ± 0.53	6.64 ± 0.53	6.93 ± 0.24	8.63 ± 0.54	29.11 ± 0.99	35.12 ± 1.96	0.49 ± 0.13
	Fanchuan	6.04 ± 0.22	6.19 ± 0.18	7.04 ± 0.20	8.66 ± 0.46	29.58 ± 0.67	35.38 ± 1.57	1.49 ± 0.07
	GIF	<u>6.69</u> ± 0.35	<u>6.73</u> ± 0.42	<u>6.97</u> ± 0.18	<u>8.64</u> ± 0.48	<u>29.38</u> ± 0.81	<u>35.22</u> ± 1.65	9.3 ± 0.5
	IDEA	5.52 ± 2.13	5.95 ± 1.56	5.78 ± 2.08	8.22 ± 0.98	25.23 ± 6.83	33.93 ± 3.44	0.49 ± 0.51
	Kookmin	6.69 ± 0.41	6.68 ± 0.47	6.97 ± 0.29	8.52 ± 0.52	29.30 ± 1.14	34.79 ± 1.85	<u>0.28</u> ± 0.04
	SCIF	6.75 ± 0.33	6.77 ± 0.40	6.95 ± 0.22	8.59 ± 0.53	29.26 ± 0.81	35.07 ± 1.88	0.24 ± 0.03
	Seif	4.99 ± 0.92	5.14 ± 0.92	5.00 ± 1.30	6.53 ± 1.55	25.35 ± 1.98	31.78 ± 2.10	0.32 ± 0.02
Sets2Sets	Original	10.20 ± 0.58	10.07 ± 0.57	8.73 ± 0.63	10.85 ± 0.90	35.40 ± 1.50	41.75 ± 1.90	92.23 ± 9.47
	Retrain	9.83 ± 0.52	9.58 ± 0.33	8.30 ± 0.60	10.20 ± 0.60	34.82 ± 1.48	40.35 ± 1.95	78.23 ± 9.78
	Fanchuan	10.75 ± 0.91	10.53 ± 0.66	9.27 ± 0.41	11.26 ± 0.57	35.77 ± 1.07	41.51 ± 1.69	0.26 ± 0.03
	Kookmin	10.18 ± 0.65	10.05 ± 0.61	8.72 ± 0.67	10.85 ± 0.95	35.29 ± 1.49	<u>41.64</u> ± 1.90	0.11 ± 0.03
	SCIF	<u>10.20</u> ± 0.64	<u>10.07</u> ± 0.60	<u>8.73</u> ± 0.65	<u>10.85</u> ± 0.93	<u>35.30</u> ± 1.44	41.67 ± 1.84	0.37 ± 0.04
	Seif	9.93 ± 0.69	9.85 ± 0.65	8.45 ± 0.86	10.67 ± 1.02	34.74 ± 1.80	41.30 ± 1.95	<u>0.12</u> ± 0.07

Table 10: Results for the Tafeng dataset, averaged over seeds.

Model	Algorithm	nDCG (%)		Recall (%)		Hit (%)		Time (min)
		@10	@20	@10	@20	@10	@20	Avg/Req
DNNTSP	Original	10.73 $\pm$ 0.21	12.13 $\pm$ 0.18	13.52 $\pm$ 0.14	17.96 $\pm$ 0.10	41.34 $\pm$ 0.44	51.47 $\pm$ 0.32	521.38 $\pm$ 37.88
	Retrain	10.79 $\pm$ 0.23	12.22 $\pm$ 0.23	13.54 $\pm$ 0.05	18.09 $\pm$ 0.09	41.49 $\pm$ 0.24	51.78 $\pm$ 0.21	504.67 $\pm$ 4.5
	CEU	10.71 $\pm$ 0.22	12.14 $\pm$ 0.21	13.51 $\pm$ 0.14	18.03 $\pm$ 0.18	41.17 $\pm$ 0.44	51.44 $\pm$ 0.45	<u>0.64</u> $\pm$ 0.14
	Fanchuan	9.70 $\pm$ 0.39	11.36 $\pm$ 0.33	12.74 $\pm$ 0.10	17.78 $\pm$ 0.12	39.54 $\pm$ 0.25	51.11 $\pm$ 0.36	7.69 $\pm$ 0.52
	GIF	10.47 $\pm$ 0.64	11.78 $\pm$ 0.50	13.11 $\pm$ 0.43	17.36 $\pm$ 0.38	40.16 $\pm$ 0.86	49.95 $\pm$ 0.80	15.17 $\pm$ 1.18
	IDEA	<u>10.70</u> $\pm$ 0.22	<u>12.14</u> $\pm$ 0.21	<u>13.50</u> $\pm$ 0.12	<u>18.03</u> $\pm$ 0.13	<u>41.12</u> $\pm$ 0.50	<u>51.44</u> $\pm$ 0.38	0.22 $\pm$ 0.05
	Kookmin	10.58 $\pm$ 0.19	11.97 $\pm$ 0.27	13.25 $\pm$ 0.38	17.67 $\pm$ 0.55	40.57 $\pm$ 1.08	50.69 $\pm$ 1.26	1.29 $\pm$ 0.09
	SCIF	10.64 $\pm$ 0.23	12.08 $\pm$ 0.20	13.48 $\pm$ 0.15	18.02 $\pm$ 0.12	41.10 $\pm$ 0.48	51.44 $\pm$ 0.37	0.74 $\pm$ 0.09
Sets2Sets	Seif	9.43 $\pm$ 1.18	10.80 $\pm$ 0.95	11.68 $\pm$ 0.92	16.01 $\pm$ 0.73	37.37 $\pm$ 2.19	47.60 $\pm$ 1.15	1.22 $\pm$ 0.1
	Original	8.63 $\pm$ 0.32	9.44 $\pm$ 0.43	9.44 $\pm$ 0.41	12.52 $\pm$ 0.81	32.82 $\pm$ 1.42	40.85 $\pm$ 1.97	213.74 $\pm$ 31.83
	Retrain	8.86 $\pm$ 0.27	9.68 $\pm$ 0.38	9.68 $\pm$ 0.36	12.80 $\pm$ 0.75	33.57 $\pm$ 0.89	41.45 $\pm$ 1.69	227.7 $\pm$ 28.41
	Fanchuan	7.32 $\pm$ 0.97	7.87 $\pm$ 0.98	7.65 $\pm$ 1.14	9.88 $\pm$ 1.25	28.71 $\pm$ 2.03	35.41 $\pm$ 2.23	0.57 $\pm$ 0.06
	Kookmin	<u>8.71</u> $\pm$ 0.36	<u>9.53</u> $\pm$ 0.47	9.52 $\pm$ 0.45	<u>12.63</u> $\pm$ 0.88	<u>33.05</u> $\pm$ 1.52	41.15 $\pm$ 2.11	<u>0.17</u> $\pm$ 0.02
	SCIF	8.69 $\pm$ 0.35	9.51 $\pm$ 0.46	<u>9.52</u> $\pm$ 0.44	12.63 $\pm$ 0.87	33.05 $\pm$ 1.53	<u>41.14</u> $\pm$ 2.10	0.51 $\pm$ 0.06
	Seif	8.77 $\pm$ 0.42	9.61 $\pm$ 0.57	9.49 $\pm$ 0.51	12.65 $\pm$ 1.03	32.97 $\pm$ 1.62	41.11 $\pm$ 2.38	0.16 $\pm$ 0.02

Table 11: Unlearning Robustness (averaged across models and datasets). For spam unlearning we report average relative performance compared to retrained models (nDCG@20). For sensitive item unlearning we report the average reduction in the percentage of users receiving sensitive items in top-20 relative to retrained models (positive values indicate successful reduction).

Algorithm	Spam		Sensitive Items		
	SBR	NBR	CF	NBR	SBR
SCIF	-3.76 $\pm$ 4.98	+1.05 $\pm$ 6.86	+1.17 $\pm$ 2.74	-10.36 $\pm$ 23.16	-6.02 $\pm$ 1.27
GIF	div.	-27.19 $\pm$ 39.97	+8.38 $\pm$ 9.33	div.	div.
CEU	-13.99 $\pm$ 18.24	-0.26 $\pm$ 8.97	+1.60 $\pm$ 2.59	-20.26 $\pm$ 27.84	-21.84 $\pm$ 0.00
IDEA	div.	-28.33 $\pm$ 37.62	+5.25 $\pm$ 0.00	-26.35 $\pm$ 36.42	+14.06 $\pm$ 0.00
Fanchuan	-11.67 $\pm$ 10.82	-5.72 $\pm$ 12.45	-3.54 $\pm$ 2.81	-10.58 $\pm$ 17.70	-9.36 $\pm$ 9.66
Kookmin	-42.34 $\pm$ 27.11	+0.49 $\pm$ 7.39	+1.08 $\pm$ 2.54	-6.84 $\pm$ 27.67	-1.74 $\pm$ 15.92
Seif	-64.62 $\pm$ 41.54	-8.06 $\pm$ 13.36	+2.10 $\pm$ 7.64	-11.63 $\pm$ 10.33	+0.58 $\pm$ 5.60

Table 12: Performance of SCIF with different values of max_norm.

max_norm	nDCG@20	Recall@20	Hit@20	Sensitive@20	Time (min)
10 ⁻²	0.0141	0.0329	0.0351	43.51	36.30
10 ⁻¹	0.0141	0.0329	0.0351	43.51	38.80
10 ⁰	0.0141	0.0330	0.0351	43.51	39.03
10 ¹	0.0141	0.0329	0.0351	43.51	36.18

Table 13: Performance of Kookmin with different reinitialization rates.

kookmin_init_rate	nDCG@20	Recall@20	Hit@20	Sensitive@20	Time (min)
10 ⁻⁵	0.0141	0.0329	0.0351	43.18	8.47
10 ⁻⁴	0.0141	0.0329	0.0351	43.18	8.49
10 ⁻³	0.0141	0.0329	0.0351	44.48	8.58
10 ⁻²	0.0138	0.0322	0.0344	43.83	9.68
5 · 10 ⁻²	0.0133	0.0307	0.0328	44.81	8.37

Table 14: Performance and time metrics for different unlearning algorithms and batch sizes on rsc15_test dataset.

Algorithm	Batch Size	Recall		nDCG		Hit		Time/Request (s)
		@10	@20	@10	@20	@10	@20	
Kookmin	1	0.3935	0.4722	0.2440	0.2639	0.3935	0.4722	0.7787
	2	0.4477	0.5257	0.2804	0.3002	0.4477	0.5257	0.3844
	4	0.4643	0.5433	0.2910	0.3110	0.4643	0.5433	0.1974
	8	0.4684	0.5486	0.2935	0.3138	0.4684	0.5486	0.1046
	16	0.4789	0.5578	0.3021	0.3221	0.4789	0.5578	0.0586
	32	0.4862	0.5662	0.3085	0.3288	0.4862	0.5662	0.0358
	64	0.4886	0.5688	0.3105	0.3309	0.4886	0.5688	0.0229
	128	0.4934	0.5719	0.3139	0.3338	0.4934	0.5719	0.0186
	256	0.4944	0.5747	0.3149	0.3353	0.4944	0.5747	0.0121
	512	0.4939	0.5742	0.3147	0.3351	0.4939	0.5742	0.0088
SCIF	1	0.0235	0.0363	0.0129	0.0161	0.0235	0.0363	13.5533
	2	0.0164	0.0287	0.0073	0.0104	0.0164	0.0287	13.8718
	4	0.4659	0.5546	0.2824	0.3049	0.4659	0.5546	13.5540
	8	0.4937	0.5767	0.3103	0.3314	0.4937	0.5767	13.1827
	16	0.4954	0.5772	0.3143	0.3352	0.4954	0.5772	13.1483
	32	0.4954	0.5776	0.3148	0.3357	0.4954	0.5776	8.6703
	64	0.4956	0.5777	0.3148	0.3357	0.4956	0.5777	4.4762
	128	0.4955	0.5773	0.3148	0.3357	0.4955	0.5773	2.3410
	256	0.4956	0.5775	0.3148	0.3357	0.4956	0.5775	1.2597
	512	0.4956	0.5775	0.3148	0.3356	0.4956	0.5775	0.7366
Fanchuan	1	0.3494	0.4271	0.2105	0.2302	0.3494	0.4271	0.9013
	2	0.3036	0.3813	0.1778	0.1975	0.3036	0.3813	0.4530
	4	0.2344	0.3032	0.1372	0.1545	0.2344	0.3032	0.3424
	8	0.2657	0.3421	0.1588	0.1781	0.2657	0.3421	0.1572
	16	0.3440	0.4223	0.2084	0.2282	0.3440	0.4223	0.0762
	32	0.4007	0.4801	0.2492	0.2694	0.4007	0.4801	0.0501
	64	0.4328	0.5113	0.2730	0.2929	0.4328	0.5113	0.0369
	128	0.4415	0.5227	0.2783	0.2989	0.4415	0.5227	0.0275
	256	0.4509	0.5291	0.2840	0.3039	0.4509	0.5291	0.0209
	512	0.4235	0.5001	0.2682	0.2877	0.4235	0.5001	0.0226
Seif	1	0.0017	0.0030	0.0009	0.0012	0.0017	0.0030	1.1711
	2	0.0005	0.0016	0.0002	0.0005	0.0005	0.0016	0.4479
	4	0.0005	0.0016	0.0002	0.0005	0.0005	0.0016	0.2781
	8	0.0014	0.0022	0.0007	0.0008	0.0014	0.0022	0.1681
	16	0.0014	0.0024	0.0009	0.0011	0.0014	0.0024	0.1211
	32	0.0009	0.0015	0.0005	0.0007	0.0009	0.0015	0.1004
	64	0.0016	0.0033	0.0008	0.0012	0.0016	0.0033	0.0667
	128	0.1032	0.1456	0.0572	0.0679	0.1032	0.1456	0.0351
	256	0.3251	0.3961	0.2064	0.2244	0.3251	0.3961	0.0199
	512	0.3748	0.4495	0.2399	0.2589	0.3748	0.4495	0.0112

Table 15: Baseline results for CF datasets.

Model	Dataset	nDCG (%)		Recall (%)		Hit (%)	
		@10	@20	@10	@20	@10	@20
ItemKNN	Food	0.41	0.49	0.70	1.00	0.75	1.06
Pop	Food	0.13	0.20	0.28	0.56	0.29	0.60
Random	Food	0.00	0.00	0.01	0.01	0.01	0.02
ItemKNN	MovieLens	7.67	9.09	6.96	12.11	32.72	45.94
Pop	MovieLens	5.66	6.44	4.85	7.96	25.26	35.39
Random	MovieLens	0.09	0.11	0.05	0.11	0.75	1.53

Table 16: Baseline results for SBR datasets.

Model	Dataset	nDCG (%)		Recall (%)		Hit (%)	
		@10	@20	@10	@20	@10	@20
Pop	30Music	1.36	1.24	0.33	0.57	7.85	11.32
Random	30Music	0.00	0.00	0.00	0.00	0.05	0.10
Pop	Books	0.27	0.36	0.48	0.83	0.52	0.90
Random	Books	0.00	0.00	0.00	0.00	0.00	0.00
SKNN	Books	1.21	1.76	3.30	5.46	3.30	5.46
Pop	NowP	1.42	1.50	2.00	2.28	4.05	4.83
Random	NowP	0.01	0.01	0.00	0.00	0.08	0.15
Pop	RSC15	1.38	1.75	2.44	3.94	2.45	3.96
Random	RSC15	0.01	0.01	0.02	0.04	0.02	0.04

Table 17: Baseline results for NBR datasets.

Model	Dataset	nDCG (%)		Recall (%)		Hit (%)	
		@10	@20	@10	@20	@10	@20
GPTopFreq	Dunnhumby	8.76	8.82	8.44	10.56	32.98	39.87
Pop	Dunnhumby	6.95	7.05	6.93	8.85	29.64	36.65
Random	Dunnhumby	0.22	0.25	0.23	0.39	1.45	2.26
TIFUKNN	Dunnhumby	3.74	3.85	3.36	4.28	13.41	16.55
GPTopFreq	Instacart	17.78	20.00	17.27	25.68	67.27	76.99
Pop	Instacart	9.86	9.95	7.29	9.97	45.89	52.48
Random	Instacart	0.13	0.18	0.12	0.26	1.06	2.20
TIFUKNN	Instacart	25.12	26.91	25.84	33.77	75.37	81.13
GPTopFreq	TaFeng	7.88	9.26	10.71	14.90	33.16	44.19
Pop	TaFeng	9.87	10.44	9.94	11.96	29.49	36.97
Random	TaFeng	0.09	0.13	0.10	0.21	0.67	1.42
TIFUKNN	TaFeng	7.94	9.76	11.29	17.07	35.79	48.54

References

- [1] Chaochao Chen, Jiaming Zhang, Yizhao Zhang, Li Zhang, Lingjuan Lyu, Yuyuan Li, Biao Gong, and Chenggang Yan. 2024. CURE4Rec: A Benchmark for Recommendation Unlearning with Deeper influence. In *NeurIPS*.
- [2] Xiang Li, Wenqi Wei, and Bhavani Thuraisingham. 2025. MUBox: A Critical Evaluation Framework of Deep Machine Unlearning [Systematization of Knowledge Paper]. In *Proceedings of the 30th ACM Symposium on Access Control Models and Technologies (USA) (SACMAT '25)*. Association for Computing Machinery, New York, NY, USA, 175–188. doi:10.1145/3734436.3734454